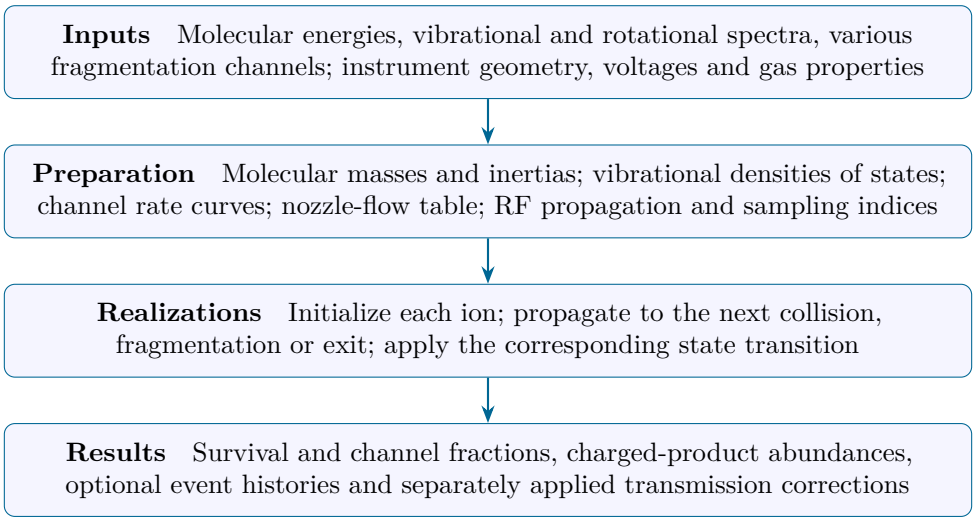
APi-TOF simulation

Model and numerical methods

A mathematical description of preprocessing, event dynamics and serial fragmentation

1 The pipeline

An atmospheric pressure interface time-of-flight (APi-TOF) mass spectrometer accelerates charged molecular clusters through residual carrier gas. Gas collisions transfer translational energy into internal excitation, enabling fragmentation. The model follows independent ion realizations through the simulated interface, not through the full flight tube or detector electronics. Its foundations are kinetic gas theory, microcanonical energy partition and phase-space detailed balance [1, 2, 3]; experimental applications include Ref. [6].



The dynamical state is

$$X = (i, \mathbf{r}, \mathbf{v}, \boldsymbol{\omega}, E_v, t), \qquad E_r = \tfrac{1}{2} I_i |\boldsymbol{\omega}|^2, \qquad E = E_v + E_r.$$

Here i identifies the current cluster, $\mathbf{r} = (x, y, z)$ is position, $\mathbf{v}$ its velocity, $\boldsymbol{\omega}$ its angular velocity, I_i its spherical moment of inertia, E_v its vibrational excitation energy and t absolute time. E_r and E are rotational and total internal energy. The axial coordinate z increases towards the downstream exit L. For each channel c, D_c is its fragmentation threshold and $k_c(E)$ its rotation-averaged fragmentation rate. The gas has molecular mass m_g, radius R_g, number density n, temperature T and axial bulk speed w. The ion has mass M_i, radius R_i and charge q_i; k_B, $\hbar$ and e denote Boltzmann's constant, reduced Planck's constant and positive elementary charge.

Runtime energies are in joules (J); velocities in $\mathrm{m\,s^{-1}}$, angular velocities in $\mathrm{s^{-1}}$, positions in metres (m) and rates in $\mathrm{s^{-1}}$. Preprocessing commonly uses the energy coordinate $\epsilon = E/k_\mathrm{B}$ in kelvin (K). A kelvin energy coordinate is not a cluster temperature. Every additional symbol is defined where introduced; Appendix A collects the notation. Pages 2–6 give the operational overview; the appendices give the derivations and numerical conventions.

2 A discrete-event simulation

The process is a piecewise deterministic Markov process: fields determine flight between random events, while collisions and fragmentation change the state. The gas is locally Maxwellian; clusters are rigid spheres for rotational and collision mechanics, with harmonic vibrational spectra for their state counts. Gas molecules have translational motion only. Collisions are binary. Internal redistribution is represented statistically after an accepted collision; E is constant during a collision-free flight. Fragmentation is irreversible along the tracked trajectory, with one charged daughter followed in a serial breakup.

1. Initialize X at $\mathbf{r} = 0, t = 0$ from the thermal laws on page 3.
2. For every accessible channel draw $T_c = -\log U_c / k_c(E)$, where the U_c are independent uniforms on $(0, 1)$; set $T_c = \infty$ when $k_c = 0$. The fragmentation deadline is $T_f = \min_c T_c$ and its channel is c_*.
3. Draw a unit exponential collision budget $B = -\log U_g$. Propagate the ion in the fields while accumulating

$$H(s) = \int_0^s \lambda\big(\mathbf{r}(t+u), \mathbf{v}(t+u)\big)\, \mathrm{d}u, \qquad T_g = \inf\{s : H(s) \geq B\},$$

   where λ is the local collision frequency. Stop at the first of T_g, T_f or downstream exit. Interfaces split flight integration without resetting the remaining budget or fragmentation deadline.
4. At a candidate gas collision, sample collision geometry and gas velocity. Apply the pinhole acceptance rule where relevant. If accepted, update momentum and vibrational energy, then redistribute internal energy. If rejected, keep the ion state at the candidate time. In either case restart the event competition.
5. At fragmentation, record c_*. Terminate in single-cluster mode, or sample the daughter state and continue on its outgoing channels in cluster-tree mode. At downstream exit, record the final cluster and finish the realization.

The collision clock is generally *not* an exponential in physical time: λ changes as the ion accelerates and crosses gas profiles. Its survival law is $\Pr(T_g > s) = e^{-H(s)}$. In contrast, each fragmentation clock is exponential between state-changing events because its rate depends on the fixed internal energy. Restarting fragmentation clocks at a rejected collision is valid by exponential memorylessness.

The total competing fragmentation rate is $K_f = \sum_c k_c(E)$:

$$\Pr(T_f > s) = e^{-K_f s}, \qquad \Pr(c_* = c \mid \text{fragmentation}, E) = \frac{k_c(E)}{K_f}.$$

The implementation draws individual channel clocks, which is equivalent to one $\mathrm{Exp}(K_f)$ clock and a categorical channel draw. A sampled fragmentation clock remains in force across chamber boundaries. The integrated-hazard solver locates gas events with bounded thinning in RF fields and adaptive quadrature elsewhere along the deterministic flight (Appendix B).

For N completed realizations, an observed fraction is $\widehat{p} = n_p/N$, where n_p counts the specified outcome. Its ordinary binomial standard-error estimate is $\sqrt{\widehat{p}(1-\widehat{p})/N}$; it describes Monte Carlo sampling uncertainty, not errors in molecular inputs or model assumptions.

3 The distributions

Let $\rho_i(u)$ be the vibrational density of states (DOS) of cluster i at excitation $u \geq 0$. A DOS has units of inverse energy; a histogram approximates it on a finite mesh. In the formulas below, all integrals use the same energy unit. Appendix C specifies the code's discrete convention.

Initial state. With chamber temperature T_0, Cartesian velocity and angular-velocity components are independent:

$$v_x, v_y \sim \mathcal{N}(0, k_B T_0/M_i), \qquad v_z \sim |\mathcal{N}(0, k_B T_0/M_i)|, \tag{1}$$

$$\omega_j \sim \mathcal{N}(0, k_B T_0/I_i), \qquad p_{v,0}(u) = \frac{\rho_i(u) e^{-u/(k_B T_0)}}{\int_0^\infty \rho_i(a) e^{-a/(k_B T_0)}\,\mathrm{d}a}. \tag{2}$$

The second parameter of $\mathcal{N}$ is variance. The axial half-normal initializes downstream motion; it is not a flux-weighted inlet distribution.

Gas collision. At a collision location define the relative ion speed $v = |\mathbf{v} - w\hat{\mathbf{z}}|$, thermal variance $a = k_B T/m_g$, effective radius $\mathcal{R} = R_i + R_g$, and polar angle $0 \leq \theta \leq \pi$ between relative ion velocity and the outward collision normal. The gas normal variable u_n is positive *inwards*, in the bulk-gas frame. The approaching normal speed is $s = u_n + v\cos\theta > 0$. Then [2]

$$\lambda(v) = \pi n \mathcal{R}^2 \left[\left(\frac{a}{v} + v \right) \operatorname{erf}\left(\frac{v}{\sqrt{2a}} \right) + \sqrt{\frac{2a}{\pi}} e^{-v^2/(2a)} \right], \tag{3}$$

$$f(\theta, u_n) = \frac{2\pi n \mathcal{R}^2}{\lambda(v)\sqrt{2\pi a}} (u_n + v\cos\theta) e^{-u_n^2/(2a)} \sin\theta, \quad u_n > -v\cos\theta. \tag{4}$$

At $v = 0$, use $\lambda(0) = 2\sqrt{2\pi a}\, n\mathcal{R}^2$. The surface azimuth ϕ and tangential direction α are independent $\mathrm{Unif}(0, 2\pi)$. Tangential gas speed u_τ is Rayleigh:

$$p(u_\tau) = \frac{u_\tau}{a} e^{-u_\tau^2/(2a)}, \qquad u_\tau \geq 0.$$

Internal-energy partition. For a specified available energy $A \geq 0$, both collision excitation and rotational–vibrational redistribution use

$$p_v(u \mid A) = \frac{\rho_i(u)\sqrt{A-u}}{\int_0^A \rho_i(b)\sqrt{A-b}\,\mathrm{d}b}, \qquad 0 \leq u < A. \tag{5}$$

For collision excitation, $A = E_{v,\mathrm{old}} + \mu_g s^2/2$ with $\mu_g = M_i m_g/(M_i + m_g)$; for internal redistribution, $A = E_v + E_r$. The factor $\sqrt{A-u}$ is the remaining three-dimensional translational or spherical-rotational DOS. It is a phase-space weight, not an assumed energy-transfer efficiency [1, 2].

Fragmentation. The clocks have densities $f_c(s) = k_c(E) e^{-k_c(E)s}$ for $s \geq 0$; $k_c(E) = 0$ for $E \leq D_c$. Their rates come from a rotation average of a zero-angular-momentum phase-space rate (Appendix D). Daughter states use a separate flux-weighted constrained distribution (Appendix G), not Eq. (5) applied independently to each product.

4 Preprocessing

Preparation separates molecular and instrument quantities from repeated random realizations. Inputs may combine electronic energies and spectra from different quantum-chemistry sources; their physical consistency remains an input requirement. In particular, the energy difference defining D_c must use the same ground-state convention as the vibrational excitation energies.

Molecular quantities. For rotational temperatures $\Theta_1, \Theta_2, \Theta_3$ (K), the principal inertias and their spherical geometric mean are

$$I_j = \frac{\hbar^2}{2k_{\mathrm{B}}\Theta_j}, \qquad I = (I_1 I_2 I_3)^{1/3}, \qquad M = A_{\mathrm{nuc}} m_p, \qquad R = \sqrt{\frac{5I}{2M}}.$$

Here A_{nuc} is the supplied nucleon-count mass parameter and m_p the proton-mass constant used by the code. An atom-like product has no vibrational or rotational structure in this model. Its translational mass remains nonzero.

Vibrational temperatures ν_j specify quantum energies $k_{\mathrm{B}}\nu_j$. A Beyer–Swinehart-type recurrence tabulates vibrational state counts per energy bin, then divides by bin width to obtain ρ_i. Concatenating the two products' frequency lists produces a combined-product DOS ρ_{DN}. The actual floor/rounding rules and ground-state treatment are in Appendix C.

Fragmentation rates. For channel c, define $\epsilon = E/k_{\mathrm{B}}$, $d_c = D_c/k_{\mathrm{B}}$ and $q = \epsilon - d_c$. The input energies in hartree are converted by

$$d_c = (E_{\mathrm{el},D} + E_{\mathrm{el},N} - E_{\mathrm{el},P})\, h_K, \qquad h_K = 3.157732 \times 10^5 \text{ K/hartree}.$$

The symbol E_{el} denotes the configured energy field, including any zero-point correction selected at ingestion. No additional correction is silently inferred. Phase-space integration and parent rotation averaging give $k_c(E)$. The intermediate parent DOS cancels analytically from its numerator before evaluation. A linear recurrence prepares the energy kernel, and shared convolutions reuse product numerators and parent normalizations across channels.

Instrument quantities. Five axial regions contain: I, the first chamber segment; II, the skimmer; III, the second chamber before the quadrupole; IV, the quadrupole; V, the segment after it. Four voltage differences and their corresponding segment lengths give constant axial electric fields outside II; the skimmer's axial electric field is zero. Region I has pressure P_1; III–V have P_2, at temperature T_0 and zero bulk gas velocity. An isentropic expanding-nozzle calculation tabulates $w(z), T(z), P(z)$ in II. The local number density is always $n(z) = P(z)/(k_{\mathrm{B}}T(z))$.

For radio-frequency (RF) quadrupole propagation, preparation stores periodic fundamental matrices for each distinct mass-dependent field coefficient. Vibrational sampling can additionally prepare a DOS cumulative sum or a log-mass tree. These indices change sampling cost, not its target law. The workflow caches DOS, rate, nozzle and RF data for reuse; cached finite energy ranges and mesh widths remain part of the numerical model.

5 Sampling and state updates

Initial state and clocks. Standard normal variates scaled by Eq. (1)–(2) initialize velocities. The vibrational sampler inverts the discrete CDF proportional to $\rho_i(E_j)e^{-E_j/(k_B T_0)}$. Independent uniforms transformed by $-\log U$ give collision hazard budgets and fragmentation waits.

Collision geometry by rejection. Change variables from u_n to the approaching speed $s = u_n + v\cos\theta$. The unnormalized joint density becomes

$$g(\theta, s) = s \exp\left[-\frac{(s - v\cos\theta)^2}{2a}\right] \sin\theta, \qquad s > 0.$$

Propose uniformly on $0 \le \theta \le \pi$ and $0 \le s \le v + B_u$, where $B_u = 5\sqrt{k_B T_0/m_g}$ is fixed at simulation setup. An envelope is

$$s_* e^{-(s_* - v)^2/(2a)}, \qquad s_* = \frac{v + \sqrt{v^2 + 4a}}{2}.$$

Accept with probability g divided by this envelope and return $u_n = s - v\cos\theta$. This is rejection sampling of a finite-domain approximation to Eq. (4); the cutoff is not infinite and is based on T_0, even when local T differs [7]. Draw $u_\tau = \sqrt{-2a\log U}$ and independent ϕ, α to complete the collision geometry.

Vibrational sampling. On the energy mesh E_j, the exact discrete target used for Eq. (5) is

$$p_j(A) = \frac{\rho_i(E_j)\sqrt{A - E_j}}{\sum_{E_\ell < A} \rho_i(E_\ell)\sqrt{A - E_\ell}}, \qquad E_j < A.$$

A direct CDF scan is the reference method. A DOS proposal with residual-energy rejection removes repeated weighted scans. A grouped log-DOS tree further bounds the rejection overhead while preserving the same $p_j(A)$. Appendix E gives both indexed samplers and their acceptance probabilities. If there are no accessible histogram states, the sampler returns zero vibrational energy and leaves the available energy in the other modes.

Collision mechanics. Rotate into a frame whose third axis is the outward surface normal. Its gas normal velocity is $-u_n$ and its tangential velocity is u_τ along the second axis. Sample new vibrational energy from $p_j(E_{v,\mathrm{old}} + \mu_g s^2/2)$. Conservation of linear momentum, energy and the no-slip tangential collision rule gives the new ion velocity and spin (Appendix F). Transform back and restore gas bulk velocity. Then sample the vibrational share of $E_v + E_r$ and rescale spin to the remaining rotational energy.

After the quadrupole, the optional pinhole rule thins candidate gas collisions according to the gas trajectory and aperture geometry; a rejected candidate leaves velocities and internal energy unchanged. This changes collision acceptance, not whether an ion is geometrically detected. Daughter recoil, spin and vibrational energy are sampled jointly as described on page 6 and in Appendix G.

6 Simulation modes and charged daughters

Single-cluster mode follows one parent with all its outgoing channels competing on each trajectory. A realization stops at its first fragmentation or at downstream exit. For N realizations, parent survival is n_{exit}/N and first-fragmentation channel fractions are n_c/N. A terminal product assignment is not a simulation of the product's subsequent flight.

Cluster-tree mode starts with the same competition but continues the charged daughter after breakup. Position and absolute time are retained; mass, inertia, radius, DOS, rates, axial acceleration and RF basis are selected for the daughter. Neutral fragments are not propagated. A graph of species and channels is expanded into execution histories, so the same species can occur at different nodes after different parent histories. Fragmentation and post-fragmentation records distinguish parent and daughter states. Terminal pathway counts label the last breakup in a history; they do not count every intermediate breakup as a separate realization.

For $P \to D{+}N$, let m_D, m_N and I_D, I_N be product masses and inertias, $\mu = m_D m_N/(m_D{+}m_N)$, $I_\Sigma = I_D{+}I_N$, $I_* = I_D I_N/I_\Sigma$, $Q = E - D_c$ and $\mathbf{J} = I_P \boldsymbol{\omega}_P$. For a unit recoil direction $\mathbf{n}$, define

$$Q_{\mathbf{n}} = Q - \frac{(\mathbf{J}\cdot\mathbf{n})^2}{2I_\Sigma}, \qquad Z(x) = \int_0^x \rho_{DN}(V)(x-V)^{5/2}\,\mathrm{d}V \quad (x>0),$$

with $Z(x) = 0$ for $x \le 0$. Sample $\mathbf{n}$ with density proportional to $Z(Q_{\mathbf{n}})$ per unit solid angle, then total product vibrational energy V with density proportional to $\rho_{DN}(V)(Q_{\mathbf{n}} - V)^{5/2}$. Draw auxiliary energies

$$(K, A, R) = (Q_{\mathbf{n}} - V)(X_K, X_A, X_R), \qquad (X_K, X_A, X_R) \sim \mathrm{Dirichlet}(1, 1, 3/2).$$

Here K is the actual relative translational energy; A, R encode coupled orbital and spin contributions, not individual product rotational energies. With $\mathbf{J}_\perp = \mathbf{J} - (\mathbf{J}\cdot\mathbf{n})\mathbf{n}$, a uniform unit vector $\mathbf{e}_\perp$ in the plane normal to $\mathbf{n}$, and an independent isotropic unit vector $\mathbf{a}$,

$$\boldsymbol{\ell} = \mathbf{J}_\perp + \sqrt{2I_\Sigma A}\,\mathbf{e}_\perp, \qquad \mathbf{s} = \sqrt{2I_* R}\,\mathbf{a}, \qquad \mathbf{j}_D = \frac{I_D}{I_\Sigma}(\mathbf{J} - \boldsymbol{\ell}) + \mathbf{s}.$$

Split V using $p(E_{v,D} = u \mid V) \propto \rho_D(u)\rho_N(V-u)$, then assign

$$\boxed{\mathbf{v}_D = \mathbf{v}_P + \frac{m_N}{m_D + m_N}\sqrt{\frac{2K}{\mu}}\mathbf{n}, \qquad \boldsymbol{\omega}_D = \frac{\mathbf{j}_D}{I_D}, \qquad E_{v,D} = u.}$$

The construction conserves energy and both momenta in each breakup. It uses unrestricted asymptotic impact parameter and spherical products; its fixed-J product states coexist with rotation-averaged event rates. That separation is an explicit approximation (Appendix G).

Pathway-at-a-time mode runs each channel independently and combines survival as $\prod_c S_c$, where S_c is the single-channel survival estimate. Its product attribution uses an independence heuristic [5]; it does not reproduce explicit competing histories in general. It is an alternative scientific mode, not an alternative collision sampler. The command-line selector is `-simulation-mode`, with values `single-cluster`, `cluster-tree` and `pathway-at-a-time`.

A Notation, units and energy conventions

Subscripts P, D, N identify parent, charged daughter and neutral product in one selected fragmentation channel. A subscript i identifies any tracked species; c identifies a channel; j indexes a Cartesian component, a vibrational mode or an energy bin, as specified locally. A superscript + means immediately after a collision. Bold symbols are vectors; a hat denotes a unit vector. The integration element $\mathrm{d}\Omega$ is solid angle. U denotes an independent uniform random variable on $(0, 1)$ each time a new draw is requested.

Symbol	Meaning	Unit
$\mathbf{r}, \mathbf{v}, t$	Ion position, laboratory velocity and absolute time	m, m/s, s
M_i, R_i, I_i, q_i	Tracked ion mass, effective radius, spherical inertia and charge	kg, m, kg m^2, C
$\boldsymbol{\omega}, \mathbf{J}$	Ion angular velocity and spin angular momentum	s^{-1}, J s
E_v, E_r, E	Vibrational, rotational and total internal energy; $E = E_v + E_r$	J
D_c, Q	Channel threshold and available product energy $E - D_c$	J
ϵ, d_c, q, h	Energy, threshold, excess energy and bin width in kelvin coordinates	K
E_j, h_J	Runtime bin centre and width; $h_J = k_\mathrm{B} h$	J
E_el, h_K	Configured molecular energy and hartree-to-kelvin conversion	hartree, K/hartree
ν_j, Θ_j	Vibrational and principal rotational temperatures	K
A_nuc, m_p	Supplied nucleon-count mass parameter and proton mass	1, kg
ρ_i, ρ_{DN}	Vibrational DOS of one species and of the product pair	J^{-1} or K^{-1}
T_0, T, P, n, w	Chamber and local temperature, pressure, number density, axial gas flow	K, K, Pa, m^{-3}, m/s
m_g, R_g, μ_g	Gas mass, radius and ion–gas reduced mass	kg, m, kg
$a, \mathcal{R}, v$	Gas velocity variance $k_\mathrm{B}T/m_g$, collision radius and relative ion speed	m^2/s^2, m, m/s
θ, ϕ, α	Surface polar angle, surface azimuth and tangent-plane direction	rad
u_n, s, u_τ	Inward gas normal variable, approaching normal speed and tangent speed	m/s
λ, H, B	Collision frequency, cumulative collision hazard and sampled hazard budget	s^{-1}, 1, 1
$k_c, k_{0,c}, K_f$	Channel rate, zero-spin channel rate and summed channel rate	s^{-1}
T_g, T_f, T_c	Waiting times to gas collision, first fragmentation and channel c	s
L, z_j, ℓ_j, V_j	Downstream exit, interfaces, segment lengths and electrode voltages	m, m, m, V
$U_\mathrm{dc}, U_\mathrm{ac}, r_0, \Omega$	Quadrupole DC/AC voltage parameters, rod half-distance and RF angular frequency	V, V, m, s^{-1}
κ, F_j, τ	Quadrupole mass coefficient, phase fundamental matrix and RF phase	V^{-1}s^{-2}, 1, rad
γ, d, β, ξ	Gas adiabatic index, nozzle throat radius, half-angle and downstream nozzle coordinate	1, m, rad, m
r_p, λ_g, Ω_p	Pinhole radius, gas mean free path and aperture solid angle	m, m, sr
m_D, m_N, μ	Product masses and product reduced mass	kg
I_Σ, I_*	Sum and reduced product inertias	kg m^2
$\mathbf{n}, \boldsymbol{\ell}, b$	Recoil direction, orbital angular momentum and impact parameter	1, J s, m
$\mathbf{j}_D, \mathbf{j}_N, \mathbf{s}$	Product spin momenta and shifted spin coordinate	J s
K, A, R, V, u	Recoil, auxiliary transverse, auxiliary spin, total product vibrational and daughter vibrational energies	J
$Q_\mathbf{n}, Z$	Direction-dependent energy budget and its phase-space normalization	J, J$^{5/2}$

The formal harmonic DOS is a measure,

$$\rho_i(E) = \sum_{\mathbf{n} \in \mathbb{N}_0^{s_i}} \delta\left(E - \sum_{j=1}^{s_i} n_j k_\mathrm{B} \nu_j \right), \tag{6}$$

where s_i is the number of retained vibrational modes, n_j an oscillator occupation and δ the Dirac delta. Smooth-DOS formulas are interpreted as finite-bin approximations at the chosen resolution. Ground-state atoms are represented by $\delta(E)$, not by a zero DOS. The molecular histogram convention used in rate preparation is specified separately below; one must not confuse omission of zero-point energy with omission of the ground-state count.

B Flight, local gas and the event clock

Let the input lengths be $(\ell_0, \ell_1, \ell_2, \ell_3, \ell_{\mathrm{sk}})$, where the first four correspond to I, III, IV, V and the last to II. The interfaces are

$$(z_0, \dots, z_5) = (0, \ell_0, \ell_0 + \ell_{\mathrm{sk}}, \ell_0 + \ell_{\mathrm{sk}} + \ell_1, \ell_0 + \ell_{\mathrm{sk}} + \ell_1 + \ell_2, L), \qquad L = \ell_0 + \ell_{\mathrm{sk}} + \ell_1 + \ell_2 + \ell_3.$$

Five input voltages $(V_0, \dots, V_4)$ give axial fields $\mathcal{E}_j = -(V_{j+1} - V_j)/\ell_j$ for $j = 0, \dots, 3$. The corresponding ion acceleration is $a_{z,j} = q_i \mathcal{E}_j / M_i$; the acceleration in II is zero. The implemented charge parameter is a sign, so $q_i = \pm e$ in axial propagation. An arbitrary charge magnitude is not inferred from a species label.

For a flight duration s within one constant axial field,

$$z(t+s) = z(t) + v_z(t)s + \tfrac{1}{2}a_z s^2, \qquad v_z(t+s) = v_z(t) + a_z s.$$

Without a transverse field, x, y drift linearly. In IV, the implemented quadrupole equations are

$$\ddot{x} = \kappa[U_{\mathrm{ac}} \cos(\Omega t) - U_{\mathrm{dc}}]x, \qquad \ddot{y} = -\kappa[U_{\mathrm{ac}} \cos(\Omega t) - U_{\mathrm{dc}}]y, \qquad \kappa = \frac{e}{M_i r_0^2}. \tag{7}$$

These are the software's voltage/sign convention; κ uses the positive elementary charge. The input voltages specify the transverse polarity. A static field is propagated analytically with sine/cosine on the focusing axis and hyperbolic sine/cosine on the defocusing axis. When $\Omega = 0$, the AC parameter contributes as a constant voltage.

For genuine RF motion use $\tau = \Omega t$ and state $\mathbf{y}_x = (x, v_x/\Omega)^T$. Its fundamental matrix satisfies

$$F_x'(\tau) = \begin{pmatrix} 0 & 1 \\ a_{\mathrm{rf}} \cos\tau - d_{\mathrm{rf}} & 0 \end{pmatrix} F_x(\tau), \quad F_x(0) = \mathrm{I}_2, \qquad a_{\mathrm{rf}} = \frac{\kappa U_{\mathrm{ac}}}{\Omega^2}, \quad d_{\mathrm{rf}} = \frac{\kappa U_{\mathrm{dc}}}{\Omega^2}.$$

F_y uses the opposite lower-left entry. I_2 is the 2×2 identity matrix. Fourth-order Runge–Kutta prepares one 2π period, using 4096 cells by default. Values and derivatives support cubic Hermite interpolation. With monodromy matrix $\mathcal{M}_j = F_j(2\pi)$,

$$F_j(2\pi n + \tau) = F_j(\tau)\mathcal{M}_j^n, \qquad 0 \le \tau < 2\pi, \quad n \in \mathbb{Z}.$$

An anchor at the current absolute phase transfers the actual position and velocity into this basis; collisions and daughter transitions do not reset the RF phase. Persisted tables are checked for conditioning, unit Wronskian and local interpolation/integration consistency.

Outside RF fields, the default solver uses embedded Gauss(7)–Kronrod(15) quadrature for H, subdividing until estimated integral error is at most $\varepsilon_a + \varepsilon_r H_{\mathrm{panel}}$. The defaults are $\varepsilon_a = 10^{-12}$ and $\varepsilon_r = 10^{-9}$, both dimensionless. It consumes accepted panels from the left until the sampled budget falls in a panel, then refines the event location. Local horizons limit accumulated hazard. Constant-rate flights use $T_g = B/\lambda$ directly. Region interfaces and skimmer table knots are integration boundaries. Traversing an interface retains the residual budget.

In RF fields the solver uses bounded thinning over windows of at most one period, truncated at interfaces or fragmentation. Cubic Bezier control points bound each interpolated velocity-matrix entry over the window, including phase wraps and the corresponding monodromy powers. Applying these bounds to the anchor coefficients and the axial endpoint-speed bound gives $v_{\max}$. The Maxwellian collision frequency is nondecreasing in ion speed, so $M = \lambda(v_{\max})$ bounds the window hazard. Exponential proposals of rate M are accepted with probability $\lambda(t)/M$; rejected proposals are not physical collisions. The window and bound are chosen independently of the proposal draw. The exponential residual can be retained at a window boundary by memorylessness, including when switching to non-RF integration. Floating-point guards enlarge the bound and evaluated envelope violations are errors.

There are no gas collisions for $z < 0$. An ion travelling upstream is propagated under the first-region axial field until return or fragmentation; this is not automatically a loss at $z = 0$. Downstream exit occurs at $z = L$. Crossing that plane is the modeled escape outcome, not an explicit aperture/rod-impact test.

B.1 Nozzle preparation and interpolation

The skimmer uses quasi-one-dimensional, steady, ideal-gas isentropic flow on the supersonic branch. Let $\xi = z - z_1$ measure distance from its throat, d be throat radius, β its cone half-angle, γ the adiabatic index, and P_0, T_0 the upstream stagnation pressure and temperature. Define

$$\mathcal{A}(\xi) = \pi[d + \xi \tan\beta]^2, \qquad w_c = \sqrt{\frac{2\gamma k_\mathrm{B} T_0}{m_g(\gamma+1)}}, \qquad w_{\max} = \sqrt{\frac{2\gamma k_\mathrm{B} T_0}{m_g(\gamma-1)}}.$$

The sonic throat condition and conservation of mass and stagnation enthalpy give the area–velocity equation

$$\left[\frac{\gamma k_\mathrm{B} T_0/m_g - (\gamma-1)w_c^2/2}{\gamma k_\mathrm{B} T_0/m_g - (\gamma-1)w^2/2}\right]^{1/(\gamma-1)} = \frac{\mathcal{A}(\xi)}{\pi d^2}\frac{w}{w_c}. \tag{8}$$

The solver searches $w_c \le w < w_{\max}$ and refines the supersonic root with secant iterations. The tabulated thermodynamic fields are

$$T = T_0 - \frac{m_g(\gamma-1)}{2\gamma k_\mathrm{B}} w^2, \qquad P = P_0\left(\frac{T}{T_0}\right)^{\gamma/(\gamma-1)}, \qquad \rho_g = \frac{m_g P_0}{k_\mathrm{B} T_0}\left(\frac{T}{T_0}\right)^{1/(\gamma-1)}.$$

ρ_g is gas mass density, distinct from vibrational DOS ρ_i. The sonic speed is $\sqrt{\gamma k_\mathrm{B} T/m_g}$. The configured `alpha_factor` gives $\beta = \pi$ `alpha_factor`, and `dc` is the throat radius despite its name.

For skimmer length ℓ_{sk} and N_s table rows, the spacing is $\Delta z = \ell_{\mathrm{sk}}/N_s$ and rows are at $\xi_j = j\Delta z$, $j = 0, \ldots, N_s - 1$. Runtime interpolation is linear between rows and constant past the last row up to the skimmer end. At that end the chamber-P_2 gas description resumes; the nozzle table is not forced to match P_2. The collision clock and collision-geometry sampler both use $n = P/(k_\mathrm{B}T)$, local thermal variance and $\mathbf{v} - w\hat{\mathbf{z}}$ in the skimmer. Outside it, they use the corresponding chamber gas state.

B.2 Collision-frequency derivation

Let the local bulk-gas velocity distribution be

$$f_g(\mathbf{u}) = (2\pi a)^{-3/2} e^{-|\mathbf{u}|^2/(2a)}, \qquad a = k_\mathrm{B}T/m_g.$$

The collision flux through a surface element is the Maxwell density times its positive approaching normal speed. Integrating the two tangential components and the surface azimuth gives Eq. (4) before normalization. Integrating $u_n > -v\cos\theta$ gives the rate per polar angle

$$\lambda_\theta = \pi n \mathcal{R}^2 \sin\theta \left[\sqrt{\frac{2a}{\pi}} e^{-v^2\cos^2\theta/(2a)} + v\cos\theta\left(1 + \operatorname{erf}\frac{v\cos\theta}{\sqrt{2a}}\right)\right]. \tag{9}$$

Integrating over $0 \le \theta \le \pi$ yields Eq. (3); $f_\theta = \lambda_\theta/\lambda$ and $f(u_n \mid \theta) = \lambda f(\theta, u_n)/\lambda_\theta$. The joint rejection sampler avoids constructing either conditional histogram. The thermal gas law at a collision is flux weighted; drawing an ordinary Maxwell velocity and an independently uniform surface point is a different law.

C Vibrational DOS and finite energy meshes

Quantum-chemistry frequencies are converted to positive vibrational temperatures ν_j and principal rotational temperatures Θ_j. The supplied configuration determines frequency scaling and which energy contributions are combined; the simulator does not impose the frequency corrections from one particular experimental paper on every dataset.

Let h be DOS bin width in K, $n_E = \lfloor \epsilon_{\max}/h \rfloor$ the number of bins, and c_m a count for $m = 0, \ldots, n_E - 1$. The current fast recurrence initializes all $c_m = 0$. For each mode ν it computes

$$f = \lfloor \nu/h \rfloor, \qquad s = \lfloor \nu/h + 1/2 \rfloor, \qquad c_f \leftarrow c_f + 1,$$

then visits bins in increasing order,

$$c_m \leftarrow c_m + c_{m-s}, \qquad m = f + 1, \ldots, n_E - 1.$$

The resulting density is $\rho_m = c_m/h$, evaluated at midpoint $\epsilon_m = h(m + 1/2)$. The frequency is rejected if $f \geq n_E$. The seed uses a floor while the recurrence shift uses nearest-integer rounding; these are the implementation's discretization rules, not an exact enumeration of the unbinned spectrum. Meaningful resolution requires h substantially smaller than the vibrational energy scales, and convergence must be checked when low-frequency modes are important.

This molecular recurrence counts excited configurations and does not explicitly seed the all-zero occupation state. In particular, a molecular DOS array should not be assumed to contain a unit ground-state count in bin zero. This convention differs from the formal complete measure in Eq. (6). For an atom-like product, its standalone preparation array is zero: it is a sentinel for absent internal structure, not a physical zero state measure. The dedicated atom-like rate branch handles the missing degrees of freedom. The combined-product table is computed by concatenating the molecular frequency lists, or by reusing the molecular table when the other product is atom-like.

For complete state measures, independent product vibrations obey

$$\rho_{DN}(V) = \int_0^V \rho_D(u)\rho_N(V - u)\,\mathrm{d}u.$$

When resolving daughter vibrations, the product measures and this combined normalization must be prepared consistently, including any discrete ground-state masses. Convolving two excited-only molecular histograms omits states with one product in its ground state and is not equivalent to concatenating their oscillator lists. At high excitation the isolated ground-state weight is usually negligible, but this is not guaranteed near threshold. The conditional daughter sampler uses compatible complete product state measures, or a single explicitly chosen consistent continuum approximation.

At runtime the DOS conversion is

$$E_m = k_{\mathrm{B}}\epsilon_m, \qquad \rho_J(E_m) = \rho_K(\epsilon_m)/k_{\mathrm{B}}.$$

Thus $\rho_J(E)\,\mathrm{d}E = \rho_K(\epsilon)\,\mathrm{d}\epsilon$. Rate values remain in s^{-1} when their horizontal energy coordinate is converted from K to J. The DOS maximum must cover both internal-energy partition requests and the parent energies used in rate preparation. For a rate grid ending at excess energy $q_{\max}$, parent DOS must extend at least through $d_c + q_{\max}$ for every included channel.

Strict out-of-range handling raises a diagnostic instead of extrapolating. For a requested energy above the rate grid, the optional permissive runtime mode temporarily uses the last rate value; it accepts that approximation only when the outcome actually fragments, and raises if collision or exit wins. No histogram overflow is evidence that a physical rate has saturated.

D Fragmentation-rate construction

All energies in this section's integrals are in J. Put $Q = E - D_c$ and define the normalized parent rotational energy distribution

$$f_r(r \mid E) = \frac{\rho_P(E-r)\sqrt{r}}{\int_0^E \rho_P(E-r')\sqrt{r'}\,\mathrm{d}r'}, \qquad 0 \le r \le E. \tag{10}$$

For two spherical molecular products, detailed balance and phase-space counting give the zero-total-angular-momentum rate [1]

$$k_{0,c}(q_J) = \frac{C_c}{\rho_P(D_c+q_J)} \int_0^{q_J} \int_0^{q_J-T} T\sqrt{S}\,\rho_{DN}(q_J-T-S)\,\mathrm{d}S\,\mathrm{d}T, \quad C_c = \frac{4\sqrt{2}\,I_*^{3/2} I_\Sigma}{\pi\hbar^6}. \tag{11}$$

Here q_J is available product energy in J, and T, S are the transformed translational and spin-energy coordinates of Eq. 39 in Ref. [1]. They are not physical gas temperature or individual product energies. $I_\Sigma = I_D + I_N$, $I_*^{-1} = I_D^{-1} + I_N^{-1}$. The formula assumes unlimited asymptotic impact parameter; the finite-capture condition has been integrated out under that approximation.

The rotation-averaged rate used for event clocks is

$$k_c(E) = \int_0^{E-D_c} f_r(r \mid E) k_{0,c}(E - D_c - r)\,\mathrm{d}r, \quad E > D_c, \qquad k_c(E) = 0 \quad (E \le D_c). \tag{12}$$

Substitution of Eq. (11) cancels the parent vibrational DOS between f_r and k_0. Equivalently,

$$k_c(E) = C_c \frac{\displaystyle\int_{r+T+S\le Q} \sqrt{r}\,T\sqrt{S}\,\rho_{DN}(Q-r-T-S)\,\mathrm{d}r\,\mathrm{d}T\,\mathrm{d}S}{\displaystyle\int_0^E \rho_P(E-r)\sqrt{r}\,\mathrm{d}r}.$$

This expression exposes the energy support and explains why the actual simulated E_r does not separately enter the hazard. Averaging over rotation is a physical coarse-graining assumption, not merely a numerical shortcut.

D.1 An atom-like neutral

The neutral contributes no spin. For zero total angular momentum, $\mathbf{j}_D = -\boldsymbol{\ell}$ and its rotational energy is $\mu b^2 K / I_D$, with K the physical relative translational energy and μ the product reduced mass. The change $T = (1 + \mu b^2/I_D)K$ and integration over $0 \le b < \infty$ give [4]

$$k_{0,c}^{\mathrm{atom}}(q_J) = \frac{I_D}{\pi\hbar^3 \rho_P(D_c+q_J)} \int_0^{q_J} T\,\rho_D(q_J - T)\,\mathrm{d}T. \tag{13}$$

The same parent rotation average, Eq. (12), then applies. Setting $I_N = 0$ in Eq. (11) is not a valid way to obtain this limit: the phase-space dimension has changed.

D.2 Analytic checks

For a continuum product DOS $\rho_{DN}(V) = A_v V^{a_v - 1}$ with $A_v > 0$, $a_v > 0$, the double-integral numerator of Eq. (11) is

$$A_v q_J^{a_v + 5/2} \frac{\Gamma(2)\Gamma(3/2)\Gamma(a_v)}{\Gamma(a_v + 7/2)}.$$

Γ is the gamma function. The atom-like integral instead equals $A_v q_J^{a_v+1}/[a_v(a_v+1)]$. These give independent power-law and normalization checks of quadrature; they refer to the unaveraged numerator, not to the complete rate after division by parent DOS and averaging over r.

D.3 Cancelled rates and shared convolutions

The implementation evaluates rate tables in kelvin coordinates. Let h be the energy spacing, $d = d_c$, $n_d = \lfloor d/h \rfloor$, $s_j = \sqrt{h(j+1/2)}$, and $t_j = h(j+1/2)$. With $q_m = h(m+1/2)$ the stored excess-energy coordinate, the shared kernel is

$$M_m = \sum_{i=0}^{m} s_i t_{m-i}.$$

Because t_j is affine, define $P_{-1} = M_{-1} = 0$ and update

$$M_m = M_{m-1} + h\big(P_{m-1} + \tfrac{1}{2}s_m\big), \qquad P_m = P_{m-1} + s_m. \tag{14}$$

This prepares all M_m in linear time. Define discrete convolution by $(f * g)_m = \sum_{j=0}^{m} f_{m-j} g_j$. Cancellation of the intermediate parent DOS in Eq. (12), before forming any per-bin k_0 ratio, gives the molecular-product rate directly:

$$k_{c,m} = C_K h^2 \frac{[\rho_{DN} * (s * M)]_m}{[\rho_P * s]_{n_d+m}}, \qquad C_K = k_{\mathrm{B}}^{7/2} C_c. \tag{15}$$

Writing $G_i = (\Theta_{i1}\Theta_{i2}\Theta_{i3})^{1/3}$ for product $i = D, N$, the implemented prefactor is

$$C_K = \frac{2k_{\mathrm{B}}^2 (I_D + I_N)}{\pi\hbar^3 (G_D + G_N)^{3/2}}.$$

No rate time unit changes when energy is expressed in K. The conversion of both DOS and integration elements is already included in C_K.

For an atom-like neutral, the cancellation and rotation average produce

$$k_{c,m}^{\mathrm{atom}} = C_{\mathrm{atom}} h \frac{[\rho_D * M]_m}{[\rho_P * s]_{n_d+m}}, \qquad C_{\mathrm{atom}} = \frac{k_{\mathrm{B}}^2 I_D}{\pi\hbar^3}.$$

The same linear recurrence $M = s * t$ serves as the atom-like kernel; the molecular case has one additional convolution with s for the coupled product spins. The common rotation-average bin-width factor has cancelled. Final parent normalizations must be finite and strictly positive. Analytic cancellation avoids dividing by individual unpopulated or very small parent DOS bins only to multiply by those bins again during rotation averaging.

The product numerator is independent of the parent and threshold offset n_d. A batch therefore computes it once for each shared combined-DOS input and product type. Parent normalizations are reused for shared parent-DOS inputs; overlapping requested offset windows are evaluated only once. Prefactors are then applied separately for each channel. Convolutions are direct finite sums, with quadratic cost in the covered range; no FFT is used, so very large high-energy DOS values cannot leak Fourier roundoff into near-threshold bins. Only the initial kernel recurrence is linear-time.

Rounding d/h downward and using midpoint factors are finite-mesh approximations. Index convolutions preserve those weights; contributing bin-centre energies do not exactly sum to the stored output bin centre. Convergence in h is essential near a sharp threshold. Rate values are linearly interpolated on the stored excess-energy grid, not between species or channels.

In preprocessing, molecular DOS, product DOS, and a rate's parent energy range must be chosen together. A cached DOS on one mesh cannot be indexed as if it were on another. Prepared input associates each table with its actual mesh; maximum energy checks precede rate computation.

E Efficient sampling of the specified laws

The energy-partition notebook illustrates these targets and rejection envelopes [9]; the algorithms below are the implemented direct, DOS-only and grouped-DOS methods. The collision-geometry rejection envelope on page 5 follows from

$$g(\theta, s) \le s e^{-(s-v)^2/(2a)} \quad (s > v), \qquad g(\theta, s) \le s \le v \quad (s \le v).$$

Maximizing the first expression yields $s_* = (v + \sqrt{v^2 + 4a})/2$ and an envelope at least v. The proposal rectangle uses the finite cutoff $v + B_u$. Its omitted tail is a numerical approximation; the analytical collision frequency still integrates the untruncated Maxwell distribution. Density and collision-radius factors cancel in rejection acceptance, but remain essential in the collision clock. The cutoff is held at its setup-temperature value; the exponential and envelope use the local gas temperature.

For energy partitions, define $d_j = \rho_i(E_j) \ge 0$ and $\mathcal{I}(A) = \{j : E_j < A,\ d_j > 0\}$. The target mass is $d_j\sqrt{A - E_j}$. All samplers below return the bin centre, not a random continuous position inside a bin, and reject energy requests beyond the prepared DOS range.

E.1 Direct CDF inversion

Compute $W(A) = \sum_{j\in\mathcal{I}(A)} d_j\sqrt{A - E_j}$, draw $Y = UW(A)$, and scan cumulative weighted masses to find the corresponding bin. Preparation is negligible; each draw scans the accessible DOS. If $W = 0$, return $E_v = 0$. The normalization is recomputed for each available energy, since its residual-energy factor changes with A.

E.2 DOS proposal and rejection

Prepare prefix sums $C_j = \sum_{\ell\le j} d_\ell$. At energy A, restrict to bins below A and draw a bin with probability proportional to d_j by binary search in the truncated prefix sum. Let E_* be the lowest nonzero accessible bin centre. Accept with probability

$$\sqrt{\frac{A - E_j}{A - E_*}}.$$

The proposal times acceptance is proportional to the required weighted mass. The prefix data can be reused for all energies. A very steep DOS can propose mostly near the upper energy boundary, where this acceptance is small.

E.3 Grouped log-DOS tree

Prepare a balanced tree whose leaf mass is d_j and whose internal mass is the sum of its children's masses. Store logarithms and combine with log-sum-exp to avoid overflow and loss of small masses. Empty nodes have log-mass $-\infty$; padding leaves are empty.

At energy A, cover accessible bins by disjoint groups g. If $E_{g,\min}$ and $E_{g,\max}$ are a group's first and last energy centres, retain a group only when

$$A - E_{g,\min} \le 2(A - E_{g,\max});$$

otherwise split it. Let D_g be its total DOS mass. Propose a group with weight $D_g\sqrt{A - E_{g,\min}}$, select a leaf inside it in proportion to d_j, and accept with probability

$$\sqrt{\frac{A - E_j}{A - E_{g,\min}}}.$$

The same target follows after cancellation of proposal weights. Each retained group has acceptance at least $1/\sqrt{2}$. Group weights are rescaled by their largest logarithm before forming a short CDF; tree descent tests the lighter child directly so its probability is not lost when the other rounds to one. This is a computational improvement over repeated scans, not a new physical energy-partition law. The two indexed choices are `dos-only` and `group-dos`; the reference choice is `direct`.

F Collision mechanics and pinhole thinning

Let C be an orthonormal matrix whose columns are the collision-frame axes expressed in the laboratory frame. Its third axis is the outward surface normal selected by θ, ϕ. A rotation α about that axis selects the tangential gas direction as the second axis. In this section only, $(v_1, v_2, v_3) = \mathsf{C}^T(\mathbf{v} - w\hat{\mathbf{z}})$ and $(\omega_1, \omega_2, \omega_3) = \mathsf{C}^T\boldsymbol{\omega}$ denote pre-collision ion components in the bulk-gas frame. Gas components are $(0, u_\tau, -u_n)$. The construction remains defined when ion relative velocity is zero by choosing a reference axis before sampling the normal.

Write $M = M_i$, $m = m_g$, $R = R_i$, $\eta = M/m$, $\chi = m/(M+m)$, $\psi = M/(M+m)$, and $\Delta E_v = E_v^+ - E_{v,\mathrm{old}}$. Sampling Eq. (5) with $A = E_{v,\mathrm{old}} + \mu_g(u_n + v\cos\theta)^2/2$ guarantees sufficient normal relative energy. Normal momentum and energy conservation give

$$v_3^+ = -\chi u_n + \psi v_3 - \sqrt{\chi^2(-u_n - v_3)^2 - \frac{2\chi}{M}\Delta E_v}. \tag{16}$$

The negative square-root branch selects separation after impact. A negative radicand is a numerical/input error, not a reason to silently discard an unphysical energy draw.

The no-slip rough-sphere rule with $I = 2MR^2/5$ gives

$$v_1^+ = \frac{-4R\omega_2 + (3+2\eta)v_1}{7+2\eta}, \qquad v_2^+ = \frac{4R\omega_1 + 4u_\tau + (3+2\eta)v_2}{7+2\eta}, \tag{17}$$

$$\omega_1^+ = \frac{(2\eta-3)\omega_1 + 10(v_2 - u_\tau)/R}{7+2\eta}, \qquad \omega_2^+ = \frac{(2\eta-3)\omega_2 - 10v_1/R}{7+2\eta}, \tag{18}$$

$$\omega_3^+ = \omega_3. \tag{19}$$

These are the tangential momentum/rotation solution, rather than specular reflection. Transform velocity by $\mathbf{v}^+ = \mathsf{C}\mathbf{v}^+ + w\hat{\mathbf{z}}$ and spin by $\boldsymbol{\omega}^+ = \mathsf{C}\boldsymbol{\omega}^+$, where the vectors on the right are the collision-frame components just computed. Then set $E = E_v^+ + I|\boldsymbol{\omega}^+|^2/2$, draw $E_v' \sim p_v(\cdot \mid E)$, and rescale

$$\boldsymbol{\omega}' = \boldsymbol{\omega}^+\sqrt{\frac{E - E_v'}{I|\boldsymbol{\omega}^+|^2/2}}.$$

This statistical internal redistribution preserves energy and spin direction, but not rigid-body spin magnitude. Vibrational angular momentum is not a separate state variable. Consequently it must not be interpreted as an explicit angular-momentum-resolved intramolecular evolution.

F.1 Pinhole correction

This rule acts only in V, with $z_4 < z < L$, when pinhole radius r_p is configured. Let $\mathbf{u}_g = \mathsf{C}(0, u_\tau, -u_n)^T$ be the sampled gas velocity. The code classifies it as aperture-directed when

$$u_{g,z} < 0, \qquad \left[x + u_{g,x}\frac{L-z}{u_{g,z}}\right]^2 + \left[y + u_{g,y}\frac{L-z}{u_{g,z}}\right]^2 < r_p^2.$$

This is the implemented signed geometric test. For such candidates the acceptance probability is

$$p_{\mathrm{acc}} = \left(1 - e^{-d_p/\lambda_g}\right)\left(1 - \frac{\Omega_p}{2\pi}\right), \quad d_p = \sqrt{x^2 + y^2 + (L-z)^2}, \quad \lambda_g = \frac{k_\mathrm{B}T_0}{4\sqrt{2}\pi R_g^2 P_2}. \tag{20}$$

Ω_p is the solid angle subtended by the aperture disk at the ion position. With $h_p = L - z$ and $c(\varphi) = r_p^2 + x^2 + y^2 - 2r_p(x\cos\varphi + y\sin\varphi)$, it is evaluated as

$$\Omega_p = \int_0^{2\pi}\left(1 - \frac{h_p}{\sqrt{c(\varphi) + h_p^2}}\right)\frac{r_p^2 - r_p(x\cos\varphi + y\sin\varphi)}{c(\varphi)}\,\mathrm{d}\varphi.$$

The code uses a 1000-panel trapezoidal rule. Other gas candidates are accepted. Rejected candidates advance time and position but apply neither collision mechanics nor internal redistribution. Candidate collision counters include these rejected events. The correction is thinning of a gas-event process and is distinct from transmission weighting of spectra.

G Charged-daughter state distribution

This distribution is enabled in cluster-tree simulations by `fragmentation_state_method = "phase-space"`. The default, `"zero"`, instead resets daughter velocity, angular velocity and vibrational energy to zero. The sampler uses ground-inclusive harmonic state counts on the configured DOS grid; the rate tables retain their existing DOS convention.

For the selected channel $P \to D + N$, evaluate the parent's state at the actual fragmentation time. Let $Q = E - D_c$, $\mathbf{V} = \mathbf{v}_P$, $\mathbf{J} = I_P \boldsymbol{\omega}_P$, $M = m_D + m_N$, $\mu = m_D m_N / M$, $I_\Sigma = I_D + I_N$ and $I_* = I_D I_N / I_\Sigma$. This section first assumes both products are molecular spherical rotors.

Let K be the physical relative translational energy, $\mathbf{n}$ the unit relative-velocity direction, $\boldsymbol{\ell}$ the orbital angular momentum, b the asymptotic impact parameter, $\mathbf{j}_D, \mathbf{j}_N$ product spin angular momenta and $V = E_{v,D} + E_{v,N}$ total product vibrational energy. Then

$$\mathbf{v}_D - \mathbf{v}_N = \sqrt{2K/\mu}\,\mathbf{n}, \qquad \boldsymbol{\ell} \cdot \mathbf{n} = 0, \qquad |\boldsymbol{\ell}| = b\sqrt{2\mu K}, \tag{21}$$

$$\mathbf{j}_D + \mathbf{j}_N + \boldsymbol{\ell} = \mathbf{J}, \qquad K + \frac{|\mathbf{j}_D|^2}{2I_D} + \frac{|\mathbf{j}_N|^2}{2I_N} + V = Q. \tag{22}$$

Laboratory centre-of-mass kinetic energy is preserved separately. It is not available to pay D_c in an isolated fragmentation event.

G.1 From detailed balance to an outgoing flux measure

Equation 35 of Ref. [1] gives the zero-total-angular-momentum rate as an integral weighted by $K\, b\, \mathrm{d}b\, \mathrm{d}K\, \mathrm{d}^3 j_D$ and the remaining vibrational DOS. The factor K contains both translational state counting and relative collision flux. Restoring recoil and impact-parameter directions and imposing a specified parent $\mathbf{J}$ yields

$$\mathrm{d}W_{\mathbf{J}} \;\propto K\, \mathrm{d}K\, b\, \mathrm{d}b\, \mathrm{d}\phi_b\, \mathrm{d}\Omega_{\mathbf{n}}\, \mathrm{d}^3 j_D\; \rho_{DN}\left(Q - K - \frac{|\mathbf{j}_D|^2}{2I_D} - \frac{|\mathbf{J} - \boldsymbol{\ell} - \mathbf{j}_D|^2}{2I_N}\right). \tag{23}$$

ϕ_b is the azimuth of $\boldsymbol{\ell}$ in the plane normal to $\mathbf{n}$, and $\mathrm{d}\Omega_{\mathbf{n}}$ is recoil solid angle. The DOS is zero for negative argument. The extension to specified nonzero $\mathbf{J}$ retains the same spherical-rotor flux measure and conservation constraints; it is not obtained by independently drawing two product spins. Angular-momentum resolved phase-space treatments of molecular evaporation provide the broader context [8].

For the unrestricted-impact-parameter model used here, $0 \le b < \infty$. Completing the spin square with

$$\mathbf{s} = \mathbf{j}_D - \frac{I_D}{I_\Sigma}(\mathbf{J} - \boldsymbol{\ell})$$

gives

$$\frac{|\mathbf{j}_D|^2}{2I_D} + \frac{|\mathbf{j}_N|^2}{2I_N} = \frac{|\mathbf{s}|^2}{2I_*} + \frac{|\mathbf{J} - \boldsymbol{\ell}|^2}{2I_\Sigma}. \tag{24}$$

For a selected $\mathbf{n}$, set $J_\parallel = \mathbf{J} \cdot \mathbf{n}$, $\mathbf{J}_\perp = \mathbf{J} - J_\parallel \mathbf{n}$. Introduce

$$A = \frac{|\boldsymbol{\ell} - \mathbf{J}_\perp|^2}{2I_\Sigma}, \quad R = \frac{|\mathbf{s}|^2}{2I_*}, \quad Q_{\mathbf{n}} = Q - \frac{J_\parallel^2}{2I_\Sigma}.$$

Since $\boldsymbol{\ell} \perp \mathbf{n}$, energy conservation is simply $K + A + R + V = Q_{\mathbf{n}}$. These coordinates absorb the coupling between product spins and orbital angular momentum.

G.2 Jacobians and normalization

At fixed K,

$$Kb\,\mathrm{d}b\,\mathrm{d}\phi_b = \frac{\mathrm{d}^2\ell}{2\mu}, \qquad \mathrm{d}^2\ell = I_\Sigma\,\mathrm{d}A\,\mathrm{d}\phi_A, \qquad \mathrm{d}^3 s \propto \sqrt{R}\,\mathrm{d}R\,\mathrm{d}\Omega_{\mathbf{s}}.$$

ϕ_A is the direction of $\boldsymbol{\ell} - \mathbf{J}_\perp$ in the normal plane; it need not equal the original impact-parameter azimuth. The linear spin shift has unit Jacobian. Up to constants that cancel in conditional probabilities, Eq. (23) becomes

$$\mathrm{d}W_{\mathbf{J}} \propto \sqrt{R}\,\rho_{DN}(Q_{\mathbf{n}} - K - A - R)\,\mathrm{d}K\,\mathrm{d}A\,\mathrm{d}R\,\mathrm{d}\Omega_{\mathbf{n}}\,\mathrm{d}\phi_A\,\mathrm{d}\Omega_{\mathbf{s}}. \tag{25}$$

The integral at fixed total vibrational energy V is

$$\begin{aligned}\int_{K,A,R\geq 0} &\sqrt{R}\,\delta(Q_{\mathbf{n}} - V - K - A - R)\,\mathrm{d}K\,\mathrm{d}A\,\mathrm{d}R \\ &= \int_0^{Q_{\mathbf{n}}-V} \sqrt{R}(Q_{\mathbf{n}} - V - R)\,\mathrm{d}R = \frac{4}{15}(Q_{\mathbf{n}} - V)^{5/2}.\end{aligned} \tag{26}$$

Define $Z(x) = \int_0^x \rho_{DN}(V)(x-V)^{5/2}\,\mathrm{d}V$ for $x > 0$, and $Z(x) = 0$ otherwise. The fully normalized direction and vibration laws are

$$p(\mathbf{n} \mid E, \mathbf{J}, c) = \frac{Z(Q_{\mathbf{n}})}{\int_{S^2} Z(Q_{\mathbf{n}'})\,\mathrm{d}\Omega_{\mathbf{n}'}}, \tag{27}$$

$$p(V \mid \mathbf{n}, E, \mathbf{J}, c) = \frac{\rho_{DN}(V)(Q_{\mathbf{n}} - V)^{5/2}}{Z(Q_{\mathbf{n}})}, \quad 0 \leq V \leq Q_{\mathbf{n}}. \tag{28}$$

If the denominator is zero, this channel has no product support under the chosen state model; no daughter draw is defined. A numerical implementation must not replace that case with arbitrary zero energies.

Direction sampling can propose an isotropic $\mathbf{n}$ and accept with $Z(Q_{\mathbf{n}})/Z(Q)$, since Z is nondecreasing and $Q_{\mathbf{n}} \leq Q$. The azimuth around $\mathbf{J}$ is uniform; only the polar angle affects acceptance. Large J can concentrate recoil near the plane normal to $\mathbf{J}$; isotropic recoil is justified only in the $J = 0$ limit of this model.

At a fixed V, put $W = Q_{\mathbf{n}} - V$. Equation (25) gives $(K, A, R)/W \sim \mathrm{Dirichlet}(1, 1, 3/2)$. One construction draws independent $G_K \sim \mathrm{Gamma}(1, 1)$, $G_A \sim \mathrm{Gamma}(1, 1)$, $G_R \sim \mathrm{Gamma}(3/2, 1)$, where the second parameter is scale, and sets $(K, A, R) = W(G_K, G_A, G_R)/(G_K + G_A + G_R)$. Uniform directions are obtained by normalizing independent three-dimensional standard normal vectors; projecting into the normal plane and normalizing gives a uniform $\mathbf{e}_\perp$.

G.3 Product reconstruction and the zero-spin limit

With independently uniform $\mathbf{e}_\perp \perp \mathbf{n}$ and isotropic $\mathbf{a}$, reconstruct

$$\boldsymbol{\ell} = \mathbf{J}_\perp + \sqrt{2I_\Sigma A}\,\mathbf{e}_\perp, \qquad \mathbf{s} = \sqrt{2I_* R}\,\mathbf{a}, \tag{29}$$

$$\mathbf{j}_D = \frac{I_D}{I_\Sigma}(\mathbf{J} - \boldsymbol{\ell}) + \mathbf{s}, \qquad \mathbf{j}_N = \frac{I_N}{I_\Sigma}(\mathbf{J} - \boldsymbol{\ell}) - \mathbf{s}. \tag{30}$$

For compatible vibrational DOS, split the product excitation using

$$p(u \mid V) = \frac{\rho_D(u)\rho_N(V-u)}{\rho_{DN}(V)}, \qquad E_{v,D} = u, \quad E_{v,N} = V - u. \tag{31}$$

With discrete levels this is a conditional state-count distribution, including any ground-state atoms; the displayed density notation includes the corresponding point masses. A finite-bin implementation must preserve the same convolution normalization and the exact nonnegative residual energy.

The laboratory assignments are

$$\mathbf{v}_D = \mathbf{V} + \frac{m_N}{M}\sqrt{2K/\mu}\,\mathbf{n}, \qquad \boldsymbol{\omega}_D = \mathbf{j}_D/I_D, \tag{32}$$

$$\mathbf{v}_N = \mathbf{V} - \frac{m_D}{M}\sqrt{2K/\mu}\,\mathbf{n}, \qquad \boldsymbol{\omega}_N = \mathbf{j}_N/I_N. \tag{33}$$

Substitution verifies

$$m_D\mathbf{v}_D + m_N\mathbf{v}_N = M\mathbf{V}, \qquad \mathbf{j}_D + \mathbf{j}_N + \boldsymbol{\ell} = \mathbf{J},$$

$$\tfrac{1}{2}m_D|\mathbf{v}_D|^2 + \tfrac{1}{2}m_N|\mathbf{v}_N|^2 + \frac{|\mathbf{j}_D|^2}{2I_D} + \frac{|\mathbf{j}_N|^2}{2I_N} + V + D_c = \tfrac{1}{2}M|\mathbf{V}|^2 + E.$$

The asymptotic velocities are applied at the breakup location, treating molecular separation as instantaneous on instrument scales. Orbital angular momentum refers to the unresolved relative separation geometry; assigning both point trajectories the same macroscopic origin does not explicitly resolve that geometry. The neutral is discarded only after constructing the conserving two-product state.

When $\mathbf{J} = 0$, $A = \mu b^2 K/I_\Sigma$. Set $T = K + A = (1 + \mu b^2/I_\Sigma)K$ and $S = R$. Integrating K from zero to T in Eq. (25) gives

$$p(T, S \mid Q, J = 0) \propto T\sqrt{S}\,\rho_{DN}(Q - T - S),$$

exactly the kernel of Eq. (11). At fixed T, $K \sim \mathrm{Unif}(0, T)$. Thus the transformed T in the rate integral is not the physical recoil energy. In this limit the implied impact-parameter density is

$$p(b) = \frac{2\mu b}{I_\Sigma}\left(1 + \frac{\mu b^2}{I_\Sigma}\right)^{-2}, \qquad b \geq 0.$$

It has a long tail. Unlimited capture is an assumption about both rate and product-state statistics, not a geometric statement that fragments separate at arbitrarily large initial distances.

G.4 Atom-like neutral and finite capture

If N is atom-like, take its vibrational state to be its ground state and $\mathbf{j}_N = 0$. The two-rotor formulas with $I_* = 0$ are singular and must be replaced by the reduced-dimensional measure. Set $I_\Sigma = I_D$ and retain

$$Q_{\mathbf{n}} = Q - \frac{(\mathbf{J}\cdot\mathbf{n})^2}{2I_D}, \qquad A = \frac{|\boldsymbol{\ell} - \mathbf{J}_\perp|^2}{2I_D}.$$

Then the reduced flux has constant mechanical measure $\mathrm{d}K\,\mathrm{d}A$ and energy support $K + A + V = Q_{\mathbf{n}}$. Its normalization is

$$Z_{\mathrm{atom}}(x) = \int_0^x \rho_D(V)(x - V)\,\mathrm{d}V.$$

Use $p(\mathbf{n}) \propto Z_{\mathrm{atom}}(Q_{\mathbf{n}})$, $p(V \mid \mathbf{n}) \propto \rho_D(V)(Q_{\mathbf{n}} - V)$ and $(K, A)/(Q_{\mathbf{n}} - V) \sim \mathrm{Dirichlet}(1, 1)$. Set

$$\boldsymbol{\ell} = \mathbf{J}_\perp + \sqrt{2I_D A}\,\mathbf{e}_\perp, \quad \mathbf{j}_D = \mathbf{J} - \boldsymbol{\ell}, \quad E_{v,D} = V.$$

The velocity formulas are unchanged. At $J = 0$, integrating over K yields the $T\rho_D(Q - T)$ kernel in Ref. [4].

A finite capture model restricts $b \le b_{\max}$, where $b_{\max}$ can depend on translational energy and product rotational state. It can be sampled by proposing a complete state from the unrestricted law and rejecting unless

$$b = \frac{|\boldsymbol{\ell}|}{\sqrt{2\mu K}} \le b_{\max}(K, \text{product state}).$$

The entire proposal, including direction and energy partition, is redrawn after rejection. The accepted distribution is the original flux restricted to the capture domain. Corresponding rates must use the same domain. The main model here uses $b_{\max} = \infty$; Ref. [1] finds negligible integrated-rate change for its example when compared with its energy-dependent ion–dipole capture formula. That example does not establish equal accuracy of recoil or spin distributions for all channels.

G.5 Relationship to the event-rate approximation

The event clock uses $k_c(E)$ averaged over parent rotation, while the daughter transition conditions on the tracked $\mathbf{J}$. This combination is a model approximation. Conservation of a sampled product state does not prove that the event hazard has the correct dependence on angular momentum. A fully angular-momentum-resolved alternative requires rates $k_c(E, J)$ obtained by normalizing the same constrained flux with the parent DOS at specified E, J.

Within the rotation-averaged construction, the latent parent rotational energy conditioned on an event is

$$p(r \mid \text{fragmentation in } c, E) = \frac{f_r(r \mid E) k_{0,c}(E - D_c - r)}{k_c(E)}.$$

Sampling r from f_r alone omits this event-rate bias. The recorded rigid-body rotational energy is not automatically a sample of that latent distribution. None of these qualifications changes the exact conservation identities of the daughter reconstruction under its chosen spherical model.

H Mode semantics, recorded outcomes and spectra

The default `single-cluster` mode groups all selected outgoing pathways by parent. Its single realization uses a common stochastic collision/energy history for every competing channel, so events are coupled through that common history even though the channel clocks are conditionally independent at fixed E. Counts are exclusive first-fragmentation outcomes plus parent exits.

In `cluster-tree`, exactly one product continues to carry the original charge. The other product is neutral, and the workflow requires a consistent nonzero charge for productive species. The current node's outgoing pathways compete using its internal energy and channel thresholds. At a charged-daughter transition, absolute time and position are preserved; the daughter sampling kernel supplies its velocity, angular velocity and vibrational energy. The next flight uses its own mass-dependent quadrupole coefficient and RF table, not the root parent's mass. RF phase is inherited through absolute time.

Execution nodes distinguish repeated appearances of the same molecular species along different histories. Optional records store initialization, pre-collision state with acceptance flag, post-collision state for accepted events, pre-fragmentation state with pathway and destination, post-fragmentation daughter state, and downstream escape. Event histories carry all transitions; aggregate pathway counts attribute a realization to its last fragmentation. The no-fragmentation category contains trajectories reaching exit as the original root. A last-fragmentation count alone does not distinguish a daughter that later exits from one whose history terminates at a channel without a continuing execution node; that distinction requires history/exit information.

For `pathway-at-a-time`, let S_c be estimated survival with only channel c active and $F_c = 1 - S_c$. The combined parent survival is $S = \prod_c S_c$. For a nonempty set A of independently fragmenting channels, the heuristic allocation to channel $c \in A$ is

$$P_c = \sum_{\substack{A \subseteq \mathcal{C} \\ c \in A}} \left[\prod_{j \in A} F_j \prod_{j \notin A} S_j \right] \frac{F_c}{\sum_{j \in A} F_j}. \tag{34}$$

$\mathcal{C}$ is the set of channels. Terms with zero denominator have zero combination probability and contribute zero. For two channels this reduces to Ref. [5]. It allocates simultaneous independent fragmentation outcomes according to marginal fragmentation fractions, rather than measured conditional event times. It should not be substituted for the competing-clock law $k_c / \sum_j k_j$ in an individual trajectory.

Species fractions are converted to charged abundances with the product charge and summed when pathways produce the same species. An optional empirical mass-dependent transmission function multiplies those abundances downstream of the physical simulation. It does not alter collision or fragmentation hazards. A physically detected spectrum additionally depends on source concentrations, selection and instrumental transmission; parent survival is not itself a detector intensity. The 2022 sulfuric-acid/dimethylamine study illustrates the distinction between simulated survival and experimental relative signals [6].

For a single terminal classifier, fractions sum to one over completed realizations. For serial histories, the number of recorded fragmentation events can exceed the number of realizations, so an event count divided by N is not automatically a mutually exclusive outcome probability. Cancelled or failed partial runs require their persisted completion/failure metadata when interpreting counts. Fixed root seed and realization index determine each trajectory's random generator; parallel scheduling does not define a physical distribution.

I Source map and verification conventions

This document describes the mathematical model using the current integrated collision hazard, prepared RF propagator, fast DOS/rate preparation and exact discrete energy-partition targets. Earlier timestep clocks, conditional gas histogram samplers and slower quadrature constructions are omitted. Independent pathway mode is included because its scientific semantics differ, not as a recommended replacement for joint competition.

Model component	Principal source location
Molecular conversion and energy mesh	`src/apitofsim.cpp`
DOS, rate prefactors, cancelled shared convolutions	`src/densityandrate.cpp`
Collision frequency and energy samplers	`src/samplers.cpp`; `src/include/samplers.tpp`
Event competition, initialization, collision update, pinhole rule	`src/mass_spec.cpp`
Integrated hazard and flight interfaces	`src/trajectory.cpp`
Prepared RF matrices and interpolation	`src/include/rf_basis.h`
Isentropic nozzle flow	`src/skimmer.cpp`
Preparation, caches and execution modes	`python/apitofsim/workflow/runners.py`
Aggregate outcomes and spectral attribution	`python/apitofsim/workflow/sql/experiment_report.sql`; `python/apitofsim/plotting/spectrogram.py`
Transmission correction	`python/apitofsim/transmission.py`

The mathematical target, physical approximation and finite-mesh realization must be distinguished in validation. Useful checks are: numerical integration of the gas angular/velocity law against the analytical collision frequency; convergence of the integrated hazard under its tolerances; invariance of RF propagation to flight segmentation; normalization of the discrete $\rho_i(E_j)\sqrt{A - E_j}$ weights for each sampler; rate-mesh convergence against the analytic power laws in Appendix D; and event-wise energy and momentum conservation in daughter reconstruction. The last check tests kinematics, not the statistical accuracy of unrestricted capture or rotation-averaged hazards.

Several transcription details deserve explicit treatment. The printed energy balance in Eq. 23 of Ref. [1] uses $E - \epsilon_r$ on its right-hand side. With the paper's definitions, its subsequent formulas require $E - D_c - \epsilon_r = \varepsilon - \epsilon_r$. Equation 39 weights $\widetilde{\varepsilon}_t\sqrt{\widetilde{\varepsilon}_r}$; neither transformed coordinate is an individual product energy. The denominator factor in Eq. 38 is squared. The atom-like note's Eq. 11 has a single integral with kernel $T\rho_D$, rather than a limit of the two-rotor spin integral. The supporting-information Eqs. S11–S13 define the total frequency and integrated collision hazard used here. These equations were checked against rendered pages, rather than relying solely on extracted PDF text.

The physical model assumes a locally equilibrated carrier gas but permits a strongly nonequilibrium ion. It omits explicit vibrational angular momentum, anisotropic rotor dynamics, molecular separation trajectories, radiative cooling, charge exchange and neutral-product flight. Molecular energies, harmonic spectra, nozzle assumptions and capture dynamics can dominate error even when numerical sampling is well converged.

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